

Chapter 6. Functions of Several Variables

Scalar fields, domains, surfaces, level sets, and data landscapes

Single-variable calculus studies functions of the form $y = f(x)$, where one real input produces one real output. Multivariable calculus studies functions such as

$$z = f(x, y), \quad w = f(x, y, z), \quad F = f(x_1, x_2, \dots, x_n).$$

The input is no longer a point on a line. It may be a point in the plane, a point in space, or a vector in a high-dimensional parameter space. The output is still a scalar: a height, temperature, density, probability, loss, likelihood, energy, cost, or risk.

This chapter develops the geometric language for these objects. We learn how to describe domains, draw surfaces, read contour maps, study level curves and level surfaces, and use lower-dimensional views to understand high-dimensional functions.

Role in the book

The next several chapters study limits, derivatives, gradients, optimization, and multiple integrals. All of those topics depend on the viewpoint developed here: a multivariable function is a scalar field over a region of space.

Learning goals

After studying this chapter, you should be able to:

1. define functions of two, three, and n variables;
2. distinguish domain, codomain, and range;
3. find domains defined by square roots, logarithms, denominators, and geometric restrictions;
4. interpret the graph of $z = f(x, y)$ as a surface above a domain in the xy -plane;
5. recognize planes, hemispheres, paraboloids, and saddle surfaces;
6. compute and interpret level curves $f(x, y) = c$;
7. interpret level surfaces $f(x, y, z) = c$;
8. use slices, contour plots, heatmaps, and surface plots to visualize functions;
9. connect multivariable functions to probability, statistics, optimization, data science, and machine learning.

1 Functions of several variables

A *function of two variables* is a rule that assigns to each input pair (x, y) in a domain $D \subseteq \mathbb{R}^2$ exactly one real number $f(x, y)$:

$$f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}.$$

The set D is the *domain*. The set of all output values is the *range*:

$$\text{Range}(f) = \{f(x, y) : (x, y) \in D\}.$$

A *function of three variables* is a rule

$$f : E \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}, \quad (x, y, z) \mapsto f(x, y, z).$$

More generally, a *function of n variables* is a rule

$$f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}, \quad x = (x_1, x_2, \dots, x_n) \mapsto f(x_1, x_2, \dots, x_n).$$

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is often called a *scalar field* because it assigns a scalar value to each point in space.

Examples from science, statistics, and AI

Function	Input	Output	Interpretation
$T(x, y)$	longitude and latitude	temperature	weather map
$p(x, y, z)$	point in space	pressure	fluid or atmosphere
$\rho(x, y, z)$	point in space	density	mass distribution
$L(\beta_0, \beta_1)$	model parameters	squared error loss	linear regression
$\ell(\theta)$	statistical parameter	log-likelihood	inference
$J(w_1, \dots, w_n)$	model weights	training objective	machine learning

Modern viewpoint

In data science and AI, the most important functions are often high-dimensional. A neural-network loss function may depend on millions or billions of parameters. We cannot draw its graph, but the ideas of domains, level sets, slices, gradients, and local approximations remain central.

2 Domains as regions

Finding the domain is often the first step in studying a multivariable function. The algebraic restrictions are familiar from single-variable calculus, but now they describe regions in the plane or space.

Expression	Restriction
$\sqrt{g(x, y)}$	$g(x, y) \geq 0$
$\ln(g(x, y))$	$g(x, y) > 0$
$\frac{1}{g(x, y)}$	$g(x, y) \neq 0$
$\arcsin(g(x, y))$	$-1 \leq g(x, y) \leq 1$

Worked example 2.1: Square root and denominator

Find the domain of

$$f(x, y) = \frac{\sqrt{y - x}}{y}.$$

The square root requires $y - x \geq 0$, so $y \geq x$. The denominator requires $y \neq 0$. Therefore

$$D = \{(x, y) \in \mathbb{R}^2 : y \geq x, y \neq 0\}.$$

This is the half-plane above the line $y = x$, with the portion of the line $y = 0$ removed.

Worked example 2.2: Parabolic domain with excluded lines

Find the domain of

$$g(x, y) = \frac{\sqrt{x^2 - y}}{x^2 - 4}.$$

The square root requires $x^2 - y \geq 0$, so $y \leq x^2$. The denominator requires $x^2 - 4 \neq 0$, so $x \neq 2$ and $x \neq -2$. Thus

$$D = \{(x, y) \in \mathbb{R}^2 : y \leq x^2, x \neq \pm 2\}.$$

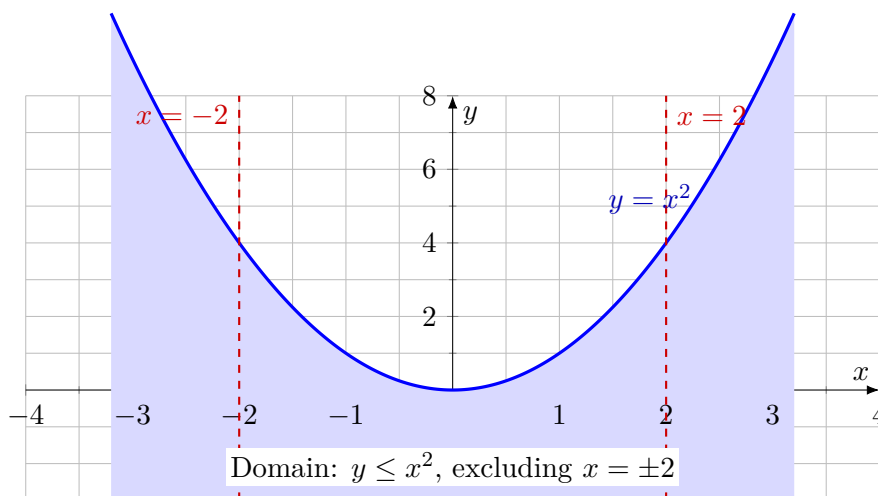


Figure 1: The domain of $\sqrt{x^2 - y}/(x^2 - 4)$ is the region below the parabola $y = x^2$, with the vertical lines $x = \pm 2$ removed.

Worked example 2.3: Logarithmic disk

Find the domain of

$$h(x, y) = \ln(9 - x^2 - y^2).$$

The logarithm requires $9 - x^2 - y^2 > 0$, or equivalently,

$$x^2 + y^2 < 9.$$

Thus the domain is the open disk of radius 3 centered at the origin.

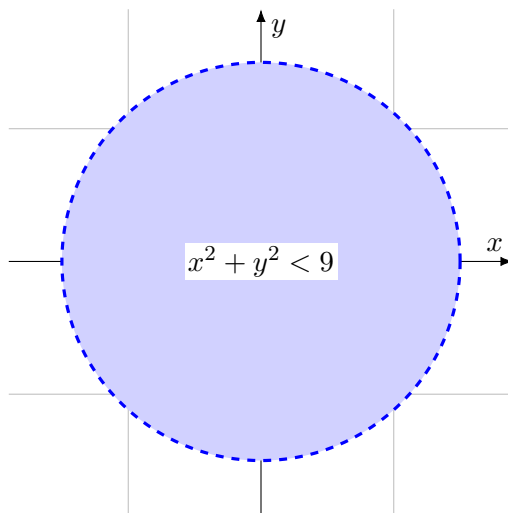


Figure 2: The domain of $\ln(9 - x^2 - y^2)$ is the open disk $x^2 + y^2 < 9$. The boundary circle is not included.

Common mistake

For $\ln(9 - x^2 - y^2)$ the boundary circle $x^2 + y^2 = 9$ is *not* included, because $\ln(0)$ is undefined.
For $\sqrt{9 - x^2 - y^2}$ the boundary is included, because $\sqrt{0} = 0$.

3 Graphs of functions of two variables

For a function $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, the *graph* is the set of points in $\mathbb{R}^3$ given by

$$\{(x, y, z) : (x, y) \in D, z = f(x, y)\}.$$

We usually write the graph as $z = f(x, y)$. The graph is a surface above or below the domain in the xy -plane.

Worked example 3.1: A plane as a graph

Let

$$f(x, y) = 6 - 2x - 3y.$$

Then

$$z = 6 - 2x - 3y,$$

or equivalently,

$$2x + 3y + z = 6.$$

This is a plane. It has x -intercept 3, y -intercept 2, and z -intercept 6.

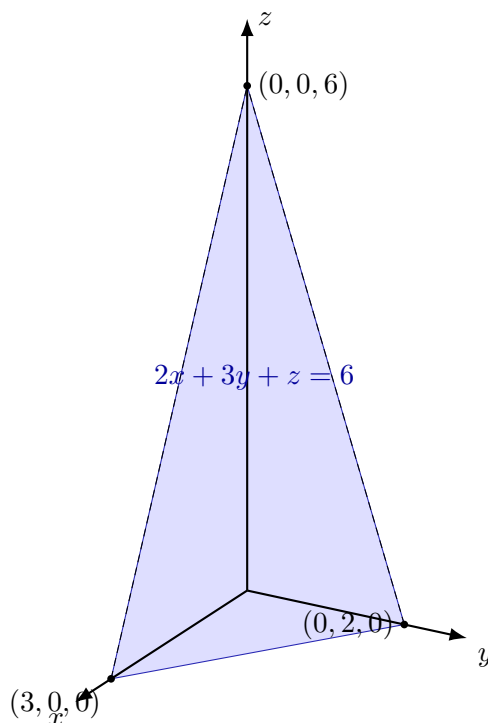


Figure 3: The graph of $f(x, y) = 6 - 2x - 3y$ is a plane. Its intercepts make the geometry transparent.

Worked example 3.2: Upper hemisphere

Let

$$f(x, y) = \sqrt{4 - x^2 - y^2}.$$

The domain is determined by $4 - x^2 - y^2 \geq 0$, so $x^2 + y^2 \leq 4$. The graph satisfies

$$z = \sqrt{4 - x^2 - y^2}.$$

Squaring both sides gives

$$x^2 + y^2 + z^2 = 4.$$

Because $z \geq 0$, the graph is the upper hemisphere of radius 2 centered at the origin. The range is $0 \leq z \leq 2$.

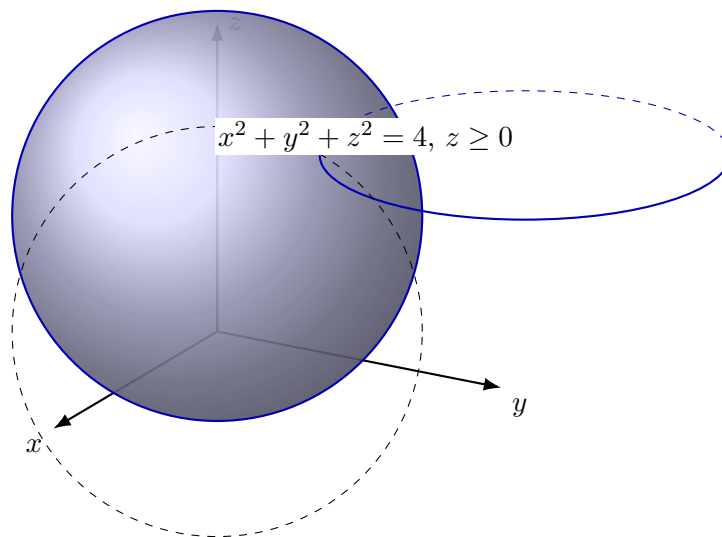


Figure 4: The graph of $f(x, y) = \sqrt{4 - x^2 - y^2}$ is the upper hemisphere of radius 2.

Worked example 3.3: Elliptic paraboloid

Let

$$f(x, y) = x^2 + 9y^2.$$

The domain is all of $\mathbb{R}^2$. Since squares are nonnegative, the range is $z \geq 0$. The graph

$$z = x^2 + 9y^2$$

is an *elliptic paraboloid*. It opens upward, and it rises faster in the y -direction than in the x -direction because the coefficient of y^2 is larger.

Worked example 3.4: A saddle surface

Let

$$f(x, y) = x^2 - y^2.$$

The graph

$$z = x^2 - y^2$$

is a saddle surface. Along the x -axis, where $y = 0$, it becomes $z = x^2$, which opens upward. Along the y -axis, where $x = 0$, it becomes $z = -y^2$, which opens downward.

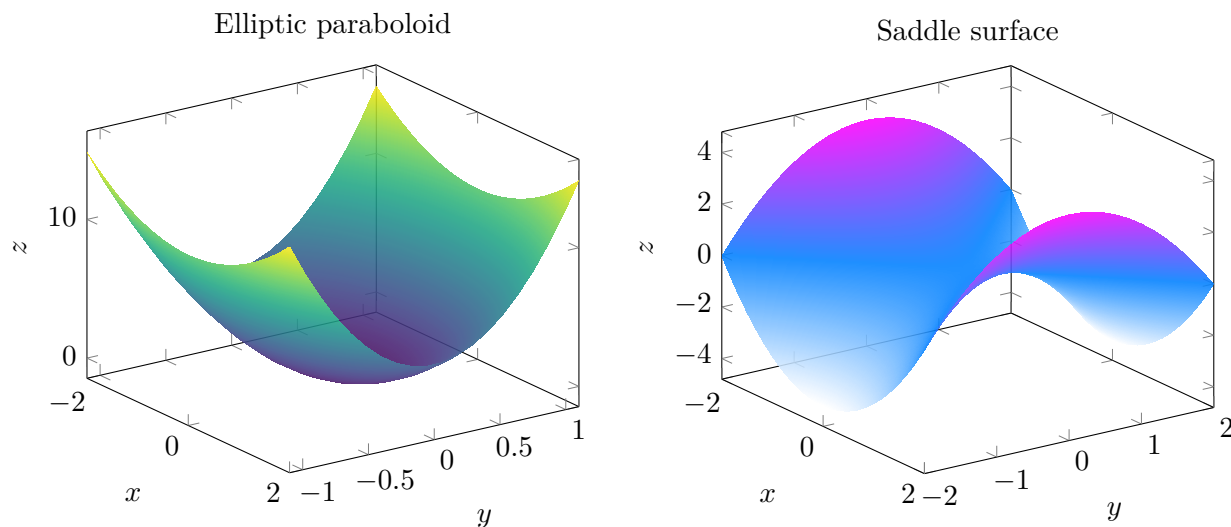


Figure 5: Two basic surfaces: the elliptic paraboloid $z = x^2 + 9y^2$ and the saddle surface $z = x^2 - y^2$.

4 Slices and traces

A surface can be understood by taking lower-dimensional slices. For $z = f(x, y)$:

- fixing $y = y_0$ gives the curve $z = f(x, y_0)$ in a vertical plane parallel to the xz -plane;
- fixing $x = x_0$ gives the curve $z = f(x_0, y)$ in a vertical plane parallel to the yz -plane;
- fixing $z = c$ gives the level curve $f(x, y) = c$ in the xy -plane.

Slices connect single-variable calculus to multivariable calculus. Partial derivatives in later chapters measure slopes of coordinate slices.

Worked example 4.1: Slices of an elliptic paraboloid

For

$$z = x^2 + 9y^2,$$

fixing $y = 0$ gives $z = x^2$, while fixing $x = 0$ gives $z = 9y^2$. The second slice is steeper.

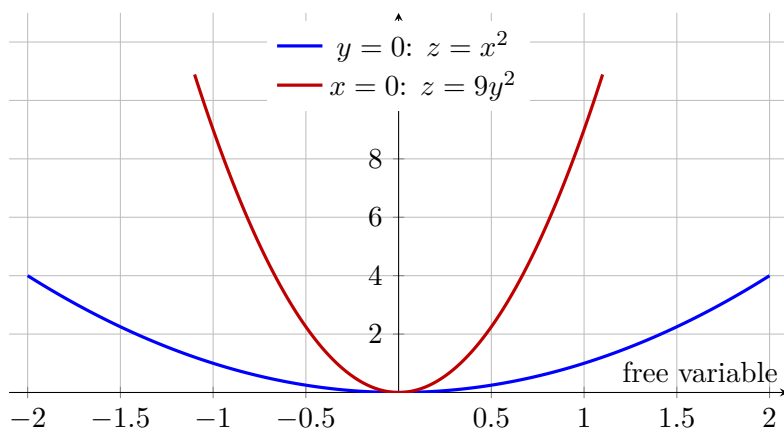


Figure 6: Coordinate slices of $z = x^2 + 9y^2$ show different rates of change in the x and y directions.

5 Level curves and contour maps

A *level curve* of a function $f(x, y)$ is a curve of the form

$$f(x, y) = c,$$

where c is a constant in the range of f .

Geometrically, the level curve $f(x, y) = c$ is obtained by intersecting the surface $z = f(x, y)$ with the horizontal plane $z = c$ and projecting the intersection down to the xy -plane. Contour maps are collections of level curves.

How to read contours

Where level curves are close together, the function changes rapidly. Where level curves are far apart, the function changes slowly.

Worked example 5.1: Circular level curves

Let

$$f(x, y) = x^2 + y^2.$$

The level curve $f(x, y) = c$ is

$$x^2 + y^2 = c.$$

For $c > 0$, this is a circle centered at the origin with radius $\sqrt{c}$.

Worked example 5.2: Elliptic level curves

Let

$$f(x, y) = x^2 + 9y^2.$$

The level curve $f(x, y) = c$ is

$$x^2 + 9y^2 = c.$$

For $c > 0$, these are ellipses.

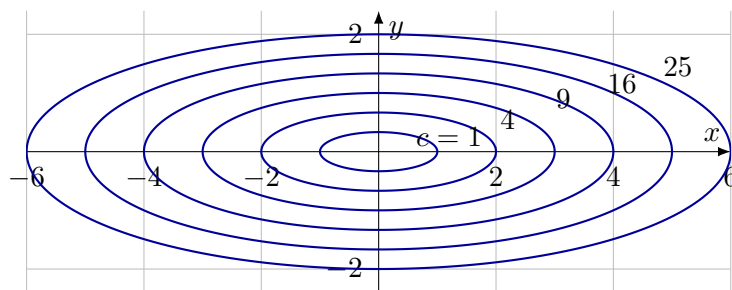


Figure 7: Level curves of $f(x, y) = x^2 + 9y^2$ are ellipses. Larger level values produce larger ellipses.

Worked example 5.3: Exponential level curves

Let

$$f(x, y) = xe^{4y}.$$

A level curve $f(x, y) = c$ satisfies

$$xe^{4y} = c.$$

If $x \neq 0$ and $c/x > 0$, then

$$y = \frac{1}{4} \ln\left(\frac{c}{x}\right).$$

This shows that level curves need not be lines, circles, parabolas, or ellipses.

6 Heatmaps and data surfaces

A multivariable function may be known through data rather than a formula. For example, a temperature field may be measured on a grid of locations, or a loss function may be evaluated at many parameter values.

A *heatmap* colors the plane according to function values. A *contour plot* draws level curves. Both are useful when a full surface plot is difficult to read.

7 Functions of three variables and level surfaces

A function of three variables has the form $f(x, y, z)$. Its graph would be $w = f(x, y, z)$, which lives in four-dimensional space. Since we cannot directly draw a four-dimensional graph, we usually study *level surfaces*:

$$f(x, y, z) = c.$$

Worked example 7.1: Spherical level surfaces

Let

$$f(x, y, z) = x^2 + y^2 + z^2.$$

The level surface $f(x, y, z) = c$ is

$$x^2 + y^2 + z^2 = c.$$

For $c > 0$, this is a sphere of radius $\sqrt{c}$ centered at the origin.

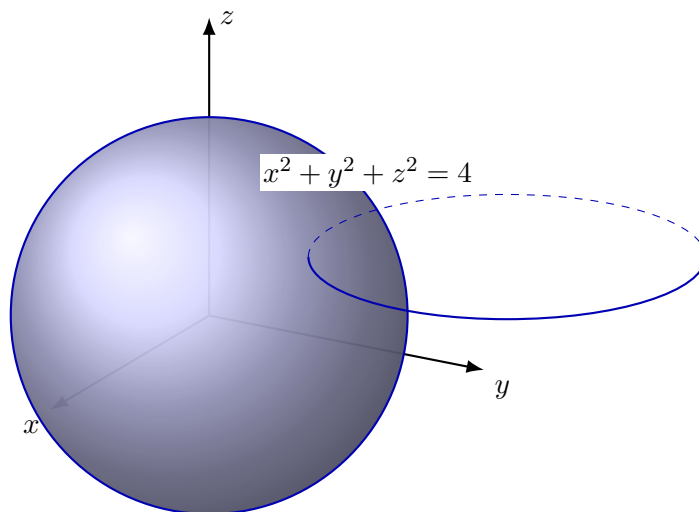


Figure 8: A level surface of $f(x, y, z) = x^2 + y^2 + z^2$ is a sphere.

Worked example 7.2: Plane level surfaces

Let

$$f(x, y, z) = 2x - y + 3z.$$

The level surface $f(x, y, z) = c$ is

$$2x - y + 3z = c.$$

For each value of c , this is a plane. Different values of c give parallel planes.

Worked example 7.3: Ellipsoid level surfaces

Let

$$f(x, y, z) = \frac{x^2}{4} + \frac{y^2}{9} + z^2.$$

The level surface $f(x, y, z) = 1$ is

$$\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1,$$

an ellipsoid with semi-axis lengths 2, 3, and 1.

8 Functions on high-dimensional spaces

In statistics and machine learning, a function often has many variables. Suppose a regression model predicts

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

The least-squares loss is

$$L(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i))^2.$$

This is a function $L : \mathbb{R}^2 \rightarrow \mathbb{R}$. For a model with p predictors, the parameter vector is

$$\beta = (\beta_0, \beta_1, \dots, \beta_p),$$

and the loss becomes a function $L : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$.

We cannot draw this graph when p is large, but we can still study slices, contours, gradients, and local quadratic approximations.

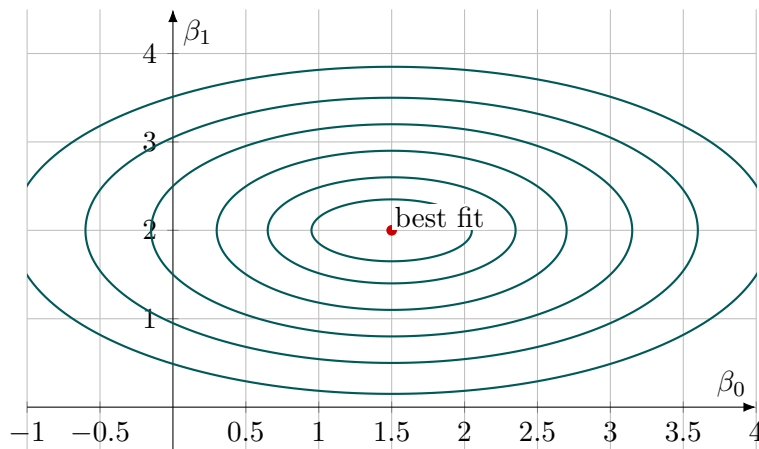


Figure 9: A two-parameter least-squares loss landscape. Smaller contours indicate lower loss.

Likelihood functions

In statistics, a likelihood function assigns to each parameter value a number measuring how compatible that parameter is with the observed data. For example, for a normal model with known variance, the log-likelihood has the form

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2 + \text{constant}.$$

For a multivariate model, θ may have many components, and the log-likelihood becomes

$$\ell(\theta) : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Thus optimization in statistics is also multivariable calculus.

9 Representations of a multivariable function

Representation	Meaning	Example
Verbal	description in words	temperature depends on longitude and latitude
Algebraic	formula	$f(x, y) = x^2 + 9y^2$
Numerical	table or data grid	measured temperatures on a map
Visual	surface, contour, heatmap	graph of $z = f(x, y)$
Algorithmic	code that computes values	simulation or machine learning model

Modern applied mathematics often moves among all five. Data are numerical, models are algebraic or algorithmic, and interpretation requires visual and verbal reasoning.

10 Common surfaces and level sets

Equation	Shape	Key feature
$z = ax + by + c$	plane	linear surface
$z = x^2 + y^2$	circular paraboloid	opens upward
$z = x^2 + 9y^2$	elliptic paraboloid	different curvature in different directions
$z = x^2 - y^2$	saddle	rises in one direction, falls in another
$x^2 + y^2 + z^2 = r^2$	sphere	level surface of distance squared
$x^2 + y^2 = r^2$	cylinder	no restriction on z
$z = \sqrt{r^2 - x^2 - y^2}$	upper hemisphere	graph over a disk
$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	ellipsoid	stretched sphere

Recognizing these shapes will help with optimization, multiple integration, and vector calculus.

11 Worked synthesis: one function, many views

Consider

$$f(x, y) = e^{-x^2 - y^2}.$$

Domain. The exponential function is defined for every real number, so the domain is all of $\mathbb{R}^2$.

Range. Since $x^2 + y^2 \geq 0$, we have $-x^2 - y^2 \leq 0$, so

$$0 < f(x, y) \leq 1.$$

The maximum value is 1 at $(0, 0)$.

Graph. The graph is a bell-shaped surface centered at the origin.

Level curves. A level curve satisfies

$$e^{-x^2 - y^2} = c.$$

For $0 < c \leq 1$, taking natural logarithms gives

$$x^2 + y^2 = -\ln c.$$

Thus the level curves are circles centered at the origin.

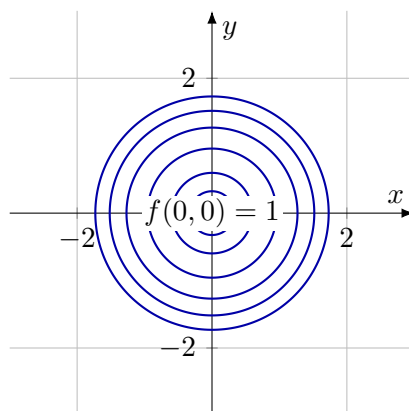


Figure 10: The function $e^{-x^2-y^2}$ has circular level curves, reflecting its radial symmetry.

This example is a first glimpse of the geometry behind multivariate normal densities, radial basis functions, kernel methods, and smooth energy landscapes.

12 Common mistakes

Mistake 1: Confusing graph and domain

The domain of $f(x, y)$ is a region in the xy -plane. The graph of f is a surface in xyz -space. For example, the domain of $\sqrt{4 - x^2 - y^2}$ is a disk, while the graph is a hemisphere.

Mistake 2: Forgetting strict inequalities

For logarithms, the input must be positive, not merely nonnegative. Thus $\ln(4 - x^2 - y^2)$ has domain $x^2 + y^2 < 4$, not $x^2 + y^2 \leq 4$.

Mistake 3: Thinking all level curves are circles

Level curves may be lines, ellipses, hyperbolas, logarithmic curves, or more complicated curves. Their shape depends on the function.

13 Practice problems

1. Find and sketch the domain of $f(x, y) = \sqrt{x + y}$.
2. Find and sketch the domain of $f(x, y) = \sqrt{x + y}/(x - y)$.
3. Find and sketch the domain of $f(x, y) = \ln(4 - x^2 - y^2)$.
4. Find the domain and range of $f(x, y) = \sqrt{9 - x^2 - y^2}$.
5. Describe the graph of $z = 3x - 2y + 5$.
6. Describe the graph of $z = x^2 + 4y^2$.

7. Describe the graph of $z = x^2 - y^2$.
8. Find the level curves of $f(x, y) = x^2 + y^2$.
9. Find the level curves of $f(x, y) = x^2 + 4y^2$.
10. Find the level curves of $f(x, y) = xy$.
11. Find the level curve of $f(x, y) = e^{-x^2-y^2}$ corresponding to $c = 1/2$.
12. Find the level surface of $f(x, y, z) = x^2 + y^2 + z^2$ through the point $(1, 2, 2)$.
13. Describe the level surfaces of $f(x, y, z) = x + y + z$.
14. Describe the level surface $x^2/4 + y^2/9 + z^2 = 1$.
15. Give an example of a function $L : \mathbb{R}^n \rightarrow \mathbb{R}$ that appears in statistics, machine learning, economics, physics, or engineering.

14 Applied and computational exercises

1. Plot the surface $z = \sin x \cos y$ on $[-2\pi, 2\pi] \times [-2\pi, 2\pi]$.
2. Draw contour plots for $f(x, y) = x^2 - y^2$.
3. Generate a grid of values for $f(x, y) = e^{-x^2-y^2}$ and display it as a heatmap.
4. For $f(x, y) = x^2 + 4y^2$, compare the contours for levels 1, 4, 9, and 16.
5. Simulate noisy data from a line $y = \beta_0 + \beta_1 x + \varepsilon$ and plot the loss contours in the (β_0, β_1) -plane.
6. Choose a function of three variables and describe at least three of its level surfaces.

15 Chapter summary

A function of several variables assigns one scalar output to each input point in a multidimensional domain:

$$f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}.$$

The main ideas of this chapter are:

- domains of multivariable functions are regions in $\mathbb{R}^n$;
- the graph of $z = f(x, y)$ is a surface in $\mathbb{R}^3$;
- level curves $f(x, y) = c$ show where a two-variable function is constant;
- level surfaces $f(x, y, z) = c$ play the same role for functions of three variables;
- slices and traces reduce multivariable functions to single-variable views;
- high-dimensional functions in statistics and AI are studied through contours, slices, heatmaps, gradients, and optimization paths.

In the next chapter, we study limits and continuity. The key new issue is that points can approach a target point from infinitely many directions.